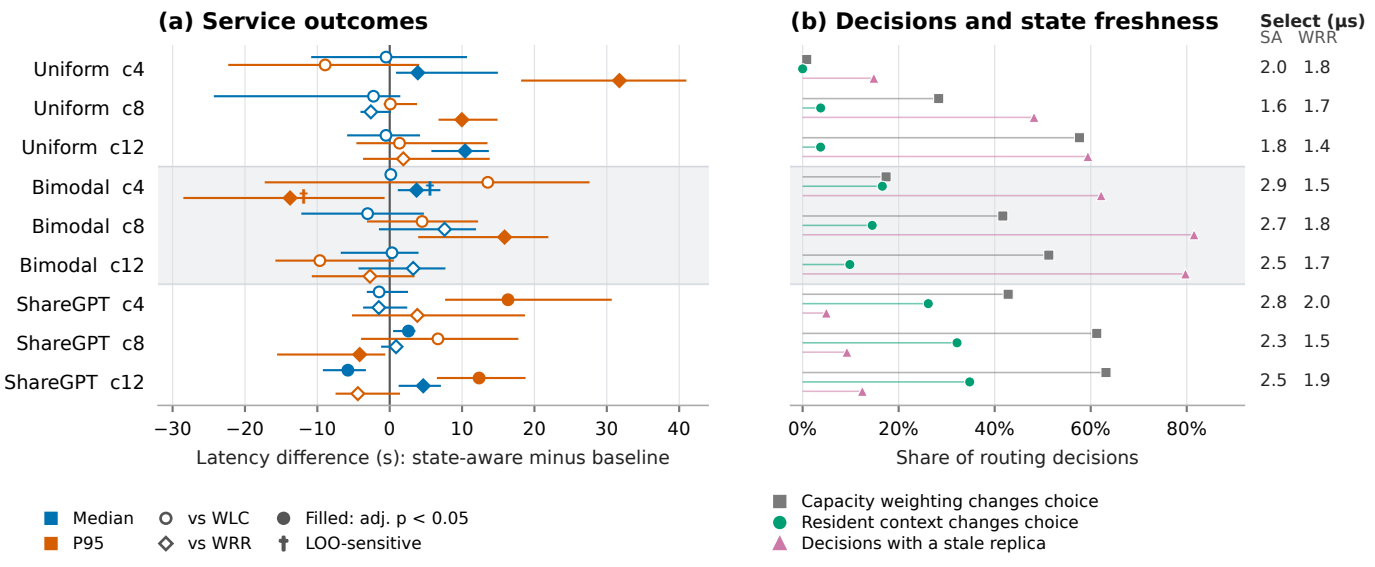
(a) Service outcomes
Uniform c4
Uniform c8
Uniform c12
Bimodal c4
Bimodal c8
Bimodal c12
ShareGPT c4
ShareGPT c8
ShareGPT c12
−30
−20
−10
0
10
20
30
40
Latency difference (s): state-aware minus baseline
Median
P95
vs WLC
vs WRR
Filled: adj. p < 0.05
LOO-sensitive
(b) Decisions and state freshness
Select (µs)
SA WRR
2.0 1.8
1.6 1.7
1.8 1.4
2.9 1.5
2.7 1.8
2.5 1.7
2.8 2.0
2.3 1.5
2.5 1.9
0%
20%
40%
60%
80%
Share of routing decisions
Capacity weighting changes choice
Resident context changes choice
Decisions with a stale replica